
CICADA 3301 · LIBER PRIMUS 0-2



SOLUTION 0-2

Volume 2

27x27 Rune Matrix · Totient Inheritance · Hidden Keys · Mobius Phase

Strongest plaintext candidate

AS I GO, THE WEATHER TURNS COLD. I MAY CRY...

A reader edition of the Volume 2 research. The wording remains close to the source while the sequence has been reorganized for clarity and continuity.

Continuing beyond COLD

Volume 1 ends at:

AS I GO, THE WEATHER TURNS COLD.

Volume 2 continues from the final state of COLD. The main new observation is that totient values appear to persist across stages of the route. In some cases, values already derived from a key or node are reused as movement distances and lead to the location of another key.

The exact rule that decides when this mechanism is activated is still unknown. However, the mechanism is visible in multiple strong examples.

Main question of Volume 2: can the same structure that produced COLD continue without changing the basic rules - and can previously derived totient values lead to the next key?

Core rules carried over from Volume 1

Mirrored node: $a-b-a \rightarrow a-\phi(b)-a$

Totent signature: $(\phi(k_1), \phi(k_2), \phi(k_3))$

Mobius phase: $p = \sum \mu(\phi(k_i)) \bmod 3$

Decryption: $\text{plaintext} = \text{ciphertext} - \text{key} \bmod 29$

The same 27x27 map

The route continues on the same 729-rune matrix. Volume 2 does not rebuild the map; it begins from the state already reached at the end of Volume 1.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27									
1	S/Z	H	E	O	G	M	IA/IO	F	1	S/Z	Y	E	NG	C/K	13	TH	J	G	AE	1	F	J	OE	1	O	N	TH	1	N	E	W	B				
2	A	S/Z	OE	EO	I	NG	U/V	1	N	AE	G	I	TH	1	H	B	N	IA/IO	1	P	B	N	1	EO	J	TH	1	A	NG	Y	X	1	B			
3	D	P	1	E	IA/IO	F	X	G	F	1	S/Z	P	T	F	P	U/V	NG	Y	NG	X	IA/IO	1	Y	H	1	B	U/V	1	G	O	P	1	NG			
4	AE	TH	1	I	AE	H	X	TH	P	G	1	Y	F	NG	C/K	J	1	TH	A	M	I	NG	I	IA/IO	Y	1	H	EO	1	T	H	AE				
5	IA/IO	X	Y	1	E	U/V	NG	1	C/K	F	N	1	EO	TH	J	I	TH	1	TH	P	1	Y	NG	EA	1	Y	N	E	Y	1	D	X	NG	W	1	
6	B	C/K	X	D	B	F	1	M	T	N	E	1	EA	J	N	L	G	B	1	X	G	TH	4	Y	I	D	A	1	NG	G	M	J				
7	R	O	L	EO	1	Y	T	P	I	TH	D	J	1	OE	H	L	1	EA	1	F	X	AE	P	G	X	1	F	L	W	EO	T	AE	1			
8	F	G	N	1	M	EO	L	N	1	NG	M	F	R	IA/IO	1	A	S/Z	1	EA	M	NG	X	X	EO	D	B	1	OE	D	M	O	1	EA			
9	EO	H	1	D	NG	G	1	OE	NG	L	U/V	R	N	T	1	AE	S/Z	Y	U/V	H	R	T	1	H	C/K	1	S/Z	OE	T	OE	NG	U/V				
10	R	1	S/Z	1	F	O	E	W	EA	OE	L	F	AE	1	M	R	NG	1	D	A	M	R	1	W	1	A	I	M	S/Z	N	1	S/Z	OE	R		
11	U/V	W	S/Z	L	IA/IO	4	OE	A	AE	NG	S/Z	D	P	T	1	C/K	R	TH	IA/IO	1	R	B	O	D	TH	F	1	Y	X	I	T	4				
12	OE	I	1	F	L	O	1	F	EA	1	R	O	OE	M	H	M	G	P	H	1	AE	H	N	O	H	Y	1	OE	S/Z	L	1	N	G	1		
13	AE	Y	1	OE	C/K	B	L	J	1	NG	L	OE	1	AE	J	EA	W	1	EA	TH	O	1	B	OE	Y	1	C/K	F	C/K	J	1	L	AE	4	N	1
14	TH	P	1	U/V	X	1	OE	X	G	1	P	F	S/Z	EO	AE	OE	1	NG	P	EO	O	E	A	1	G	AE	IA/IO	NG	TH	O	1	P	A			
15	OE	TH	R	NG	AE	1	C/K	S/Z	B	EO	Y	H	1	T	X	E	L	R	4	G	W	Y	1	J	F	I	N	IA/IO	C/K	Y	EA					
16	I	IA/IO	1	O	TH	1	E	C/K	AE	C/K	X	IA/IO	EA	1	O	L	C/K	1	F	R	D	NG	E	U/V	1	D	C/K	L	EA	S/Z	X	C/K				
17	G	IA/IO	4	W	S/Z	TH	1	F	D	NG	1	I	IA/IO	M	A	AE	G	1	W	S/Z	1	N	D	1	C/K	P	TH	X	P	EA	EA	1	W	U/V	1	
18	EA	W	1	F	W	J	Y	1	X	D	W	C/K	G	C/K	OE	1	D	X	OE	1	R	IA/IO	G	1	N	P	A	Y	P	1	C/K	Y	H	1		
19	F	E	J	EA	N	1	OE	AE	1	U/V	A	1	H	R	1	E	EA	TH	F	J	A	1	L	X	S/Z	T	1	M	F	L	F	T	1	G		
20	I	TH	1	U/V	L	G	1	X	EA	T	S/Z	L	J	P	1	L	X	I	OE	M	1	U/V	4	H	T	1	B	EO	L	D	H	B	M	1		
21	EA	R	B	1	N	H	B	E	G	EO	1	D	L	W	EO	IA/IO	P	O	1	H	E	EA	1	W	I	R	I	H	1	U/V	TH	H				
22	Y	1	N	X	B	G	J	P	U/V	1	NG	EA	F	N	I	E	D	IA/IO	NG	R	1	D	B	J	IA/IO	OE	M	I	1	F	T	1				
23	J	D	I	TH	R	L	S/Z	1	E	EO	O	G	B	T	D	1	TH	A	N	C/K	Y	1	IA/IO	S/Z	TH	D	1	NG	F	L	E					
24	G	H	C/K	1	E	O	I	T	N	X	1	P	T	F	H	R	D	E	F	T	1	AE	W	H	1	B	C/K	1	F	1	P	A	L			
25	U/V	EA	N	L	J	1	J	C/K	L	W	EA	D	U/V	D	EO	1	EA	X	D	W	H	EA	1	TH	IA/IO	AE	C/K	L	T	W						
26	E	I	C/K	1	P	OE	D	C/K	1	N	H	A	1	R	I	G	TH	EA	E	T	G	1	TH	H	O	IA/IO	W	AE	J	E	1	NG				
27	EA	D	1	O	AE	1	A	L	1	B	J	C/K	U/V	X	T	A	B	J	P	1	F	EA	1	H	D	N	IA/IO	U/V	P	S/Z	U/V	W	13			

Original working grid retained. Volume 2 focuses on the post-COLD structures highlighted in this same map.

Two ways keys appear

The current analysis suggests two different key mechanisms.

Type A - directly generated keys. A mirrored three-rune node directly generates a key by transforming its center with phi. The familiar example is H-TH-H -> H-U-H.

Type B - hidden keys. Some keys are not generated at the current node. Instead, specific totient values derived earlier in the route are reused as movement distances. Those movements lead to the location of another key in the 27x27 grid.

earlier totient values -> movement -> hidden key location

A second distinction appears once the hidden structure is found:

Non-mirrored: use the discovered key directly.

Mirrored: apply phi to the center first, then use the transformed structure as the actual key.

H-NG-C -> H-NG-C NG-B-NG -> phi(B)=T -> NG-T-NG

The activation rule is not yet fully known. What is observable is that the same relationship between symmetry and totient transformation continues to repeat.

Frozen state at the end of COLD

Volume 1 derives the COLD key from H-TH-H -> H-U-H, because $\phi(\text{TH}=2)=1=U$.

The totient signature of H-U-H is:

4-1-4

COLD ends on the ciphertext cell $A(11,7)$. So the post-COLD state contains the preserved parent H-TH-H, the generated key H-U-H, the inherited signal 4-1-4, and the endpoint $A(11,7)$.

A structural fingerprint at the endpoint

The final A is immediately inside:

$EA(10,7) - A(11,7) - EA(12,7)$

Its totient signature is:

$\phi(EA=28)=12 \quad \phi(A=24)=8 \quad \phi(EA=28)=12 \quad EA-A-EA \rightarrow 12-8-12$

This becomes important because the later I MAY key has the exact same signature.

4-I-4 locates B

Starting from the final A of COLD, reuse the inherited values from H-U-H -> 4-I-4.

D4: A(11,7) -> S(15,7) R1: S(15,7) -> B(15,8)

So the route is:

A(11,7) -> D4 -> S(15,7) -> R1 -> B(15,8)

	EO	L	N	NG	M	F
G	G	OE	NG	L	U/V	R
	E	W	EA	OE	L	F
	IA/IO	OE	A	AE	NG	S/Z
	O	F	EA	R	O	OE
K	B	L	J	NG	L	OE
	OE	X	G	P	F	S/Z
G	AE	C/K	S/Z	B	EO	Y
I	E	C/K	AE	C/K	X	IA/IO
Z	TH	F	D	NG	I	IA/IO

Original map crop around the post-COLD route. The final A, the downward branch, and B are all part of the same working grid.

B reveals NG-B-NG

B(15,8) is the center of the mirrored structure:

```
NG(13,8) - B(15,8) - NG(17,8)
```

Because the discovered structure is mirrored, the center is transformed:

```
phi(B=17)=16=T NG-B-NG -> NG-T-NG
```

The actual key is therefore NG-T-NG. Its totient signature is:

```
phi(NG=21)=12 phi(T=16)=8 phi(NG=21)=12 NG-T-NG -> 12-8-12
```

This gives the exact structural match:

```
EA-A-EA -> 12-8-12 <- NG-T-NG
```

The endpoint of COLD already contains the numerical fingerprint of the key later used for I MAY.

IO	OE	A	AE	NG
1	F	EA	1	R
	L	J	1	NG
E	X	G	1	P
E	1	C/K	S/Z	B
	C/K	AE	C/K	X
H	1	F	D	NG

B at the center of the hidden NG-B-NG structure.

Mobius fixes phase 0

For the new key NG-T-NG, the totient signature is 12-8-12.

$$\mu(12)=0 \quad \mu(8)=0 \quad \mu(12)=0 \quad p = 0$$

The key is not rotated. Importantly, the phase is fixed numerically before judging which result looks like English.

phase	result
0	I M A Y
1	S X A T H
2	I X F Y

The Mobius rule independently selects the only phase that produces I MAY.

Back to the preserved H-TH-H structure

The newly found key is then applied back to the same preserved parent structure H-TH-H - the structure that previously generated H-U-H for COLD.

Its center is TH(26,19). Reading left gives:

TH(26,19) - G(26,18) - T(26,17) - E(26,16)

Ciphertext: TH-G-T-E Repeating key: NG-T-NG-NG

TH(2) - NG(21) mod 29 = 10 = I G(6) - T(16) mod 29 = 19 = M T(16) - NG(21) mod 29 = 24 = A E(18) - NG(21) mod 29 = 26 = Y

I MAY

D	1	TH	A	N	C/K	Y	1	IA
R	D	E	F	T	1	AE	V	
EO	1	EA	X	D	W	H	E	
TH	EA	E	T	G	1	TH	I	

Original map crop: the preserved H-TH-H structure and the TH-G-T-E ciphertext line used for I MAY.

The I MAY chain

H-TH-H | $\phi(\text{TH})=\text{U}$ v H-U-H | totient signature v 4-1-4 | after COLD: D4, R1 v
B(15,8) v NG-B-NG | $\phi(\text{B})=\text{T}$ v NG-T-NG | signature 12-8-12 | Mobius phase 0 v
return to H-TH-H | read TH-G-T-E v I MAY

Strong structural cross-check: EA-A-EA $\rightarrow$ 12-8-12 $\leftarrow$ NG-T-NG. The same signature is visible at the endpoint of COLD and in the key that decrypts I MAY.

The final E opens the next node

The ciphertext for I MAY ends at E(26,16). That same rune is immediately part of:

$E(24,16) - X(25,16) - E(26,16)$

So the next mirrored structure is E-X-E. This repeats an important behavior of the model: the end of one read can become part of the next structural node.

Because the node is mirrored, transform its center:

$\text{phi}(X=14)=6=G$ E-X-E -> E-G-E

The new totient signature is:

$\text{phi}(E=18)=6$ $\text{phi}(G=6)=2$ $\text{phi}(E=18)=6$ E-G-E -> 6-2-6

X	E	L	R 4	G	V
L	C/K 1	F	R	D	N
AE	G 1	W	S/Z 1	N	I
OE 1	D	X	OE 1	R	IA/
EA	TH	F	J	A 1	I
L	X	I	OE	M 1	U/
EO	IA/IO	P	O 1	H	E
I	E	D	IA/IO	NG	F
T	D 1	TH	A	N	C/
H	R	D	E	F	T
D	EO 1	EA	X	D	V
G	TH	EA	E	T	C

Original map crop showing the vertical relation between the E-X-E region and the later J-OE-S read.

Mobius fixes phase 1

For E-G-E -> 6-2-6:

$$\mu(6)=+1 \quad \mu(2)=-1 \quad \mu(6)=+1 \quad p = (+1 - 1 + 1) \bmod 3 = 1$$

The active key becomes G-E-E. Again, the phase is determined before evaluating the plaintext.

phase	result
0	O E T Y
1	C R Y
2	O E R N

The Mobius rule independently selects phase 1, which produces CRY.

The inherited 6 leads to J

Using the outer totient value 6, move upward from the center X:

$$X(25, 16) \rightarrow U6 \rightarrow J(19, 16)$$

J is exactly the center of:

$$OE(18, 16) - J(19, 16) - OE(20, 16)$$

Reading upward from J gives the ciphertext:

$$J(19, 16) - OE(18, 16) - S(17, 16) \quad J-OE-S$$

With the active key G-E-E:

$$J(11) - G(6) = 5 = C \quad OE(22) - E(18) = 4 = R \quad S(15) - E(18) \bmod 29 = 26 = Y$$

CRY

The CRY chain

I MAY | final ciphertext rune E v E-X-E | $\phi(X)=G$ v E-G-E | totient signature v
6-2-6 | Mobius phase 1 v G-E-E | U6 from X v J(19,16) | read J-0E-S upward v CRY

The strongest proposed continuation is now:

**AS I GO, THE WEATHER TURNS COLD. I MAY
CRY...**

Totient values do not always disappear

The strongest broader observation of Volume 2 is that a totient result may remain relevant after its first use. A value can appear first as part of a key signature and later reappear as a distance or structural relation elsewhere in the grid.

In the current model, totients can participate in several roles:

1. transforming the center of a mirrored node;
2. defining a key signature;
3. determining Mobius phase;
4. acting as movement distances;
5. leading to hidden key locations.

This behavior is referred to here as totient inheritance or a totient chain. This does not mean that every earlier totient value is always reused. The exact activation rule is still unknown. What is observable is that specific derived values recur at structurally important transitions.

H-NG-C and NG-B-NG

H-NG-C is the non-mirrored example. Previously derived values are reused as distances in the grid: one value locates NG; another defines the vertical distance to H and C. The resulting H-NG-C key is non-mirrored and is therefore used directly.

NG-B-NG is the mirrored example. After COLD, H-U-H $\rightarrow$ 4-1-4 leads A $\rightarrow$ D4 $\rightarrow$ R1 $\rightarrow$ B. B is the center of NG-B-NG, and because the structure is mirrored, $\phi(B)=T$ gives NG-T-NG.

The new key is then applied back to the preserved H-TH-H structure and produces I MAY.

Y	H	1	B	U/V	1	G
I	IA/IO	Y	1	H	EO	1
Y	N	E	Y	1	D	
Y	I	D	A	1	NG	
G	X	1	F	L	W	
EO	D	B	1	OE	D	
H	C/K	1	S/Z	OE	T	
A	I	M	S/Z	N	1	
O	D	TH	F	1	Y	
O	H	Y	1	OE	S/Z	
Y	1	C/K	F	C/K	J	1
G	AE	IA/IO	NG	TH		
J	F	I	N	IA/IO		
U/V	1	D	C/K	L	EA	
P	TH	X	P	EA		
N	P	A	Y	P	1	
S/Z	T	1	M	F	L	
T	1	B	EO	L	D	
W	I	R	I	H	1	
B	J	IA/IO	OE	M		
IA/IO	S/Z	TH	D	1	NG	
W	H	1	B	C/K	1	F

Original map crop of the H-NG-C hidden-key geometry.

Current rule summary

```
mirrored 3-rune node | v apply phi to center | v key | +--> totient signature | | |  
+--> Mobius phase | | | +--> possible inherited movement values | v decrypt  
ciphertext | v next structural location
```

A second mechanism can occur:

```
earlier derived totient values | v movement through 27x27 grid | v hidden key /  
hidden node
```

If the discovered structure is mirrored, transform its center with phi. If it is non-mirrored, use it directly.

What is strongest – and what is still open

The strongest elements of Volume 2 are:

H-U-H -> 4-1-4 is inherited from COLD. D4 -> R1 reaches B at the center of NG-B-NG. NG-B-NG -> NG-T-NG follows the same mirrored-center rule. EA-A-EA and NG-T-NG share 12-8-12. Mobius fixes NG-T-NG to phase 0. TH-G-T-E decrypts exactly to I MAY. The final E of I MAY immediately opens E-X-E. E-X-E -> E-G-E follows the same center-totient rule. Mobius fixes E-G-E to phase 1. U6 lands exactly on J, another mirrored center. J-OE-S decrypts exactly to CRY.

The arithmetic and geometry are reproducible, but the complete state machine is not yet fully formalized. The remaining questions are mainly about when hidden-key search activates, which direction a movement takes, and how parent-state return should be represented as a deterministic rule.

The strongest result of Volume 2

After repeated testing and attempts to break the structure, the strongest continuation found so far remains:

AS I GO, THE WEATHER TURNS COLD. I MAY CRY...

I MAY is currently the strongest post-COLD result. CRY is the strongest continuation found after I MAY. Later candidate branches remain experimental and are not included as fixed plaintext in this volume.

Volume 2 suggests that Euler totients are doing more than transforming individual rune nodes. They appear to connect mirror symmetry, key generation, numerical signatures, Mobius phases, grid movement, hidden-key locations, and structural inheritance between stages.

The most important new result is therefore not only I MAY CRY, but the possibility that totient values form a persistent control layer governing both the cryptographic and spatial behavior of the 27x27 grid.

Reader edition · rebuilt from the Volume 2 source text and the original working map.